

Name:

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45

TRANSPORT PHENOMENA I, FALL 2025

## EXAM 3

Closed Book (Refer to the provided equations sheet)

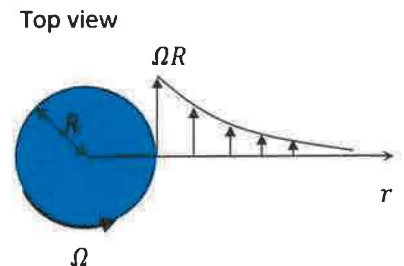
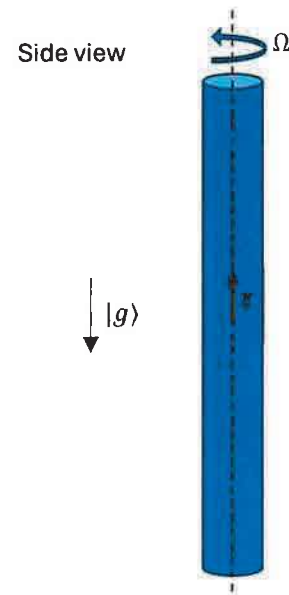
Time: 90 minutes

1. A long vertical cylinder with radius  $R$  is rotating with constant angular velocity  $\Omega$  within a large body of fluid.

Postulates:

- Steady state, laminar flow.
- Newtonian fluid with constant viscosity  $\mu$  and constant density  $\rho$ .
- Body of fluid around the cylinder is infinitely large.
- $v_r = 0$  and  $v_z = 0$
- Pressure within the fluid is not a function of  $\theta$ .

Answer the following questions in terms of parameters:  $\rho$ ,  $\mu$ ,  $\Omega$ , and  $R$ .



5/10

- 1.1. Using the mass balance (equation of continuity), demonstrate that  $v_\theta$  is not a function of  $\theta$ .

15/40

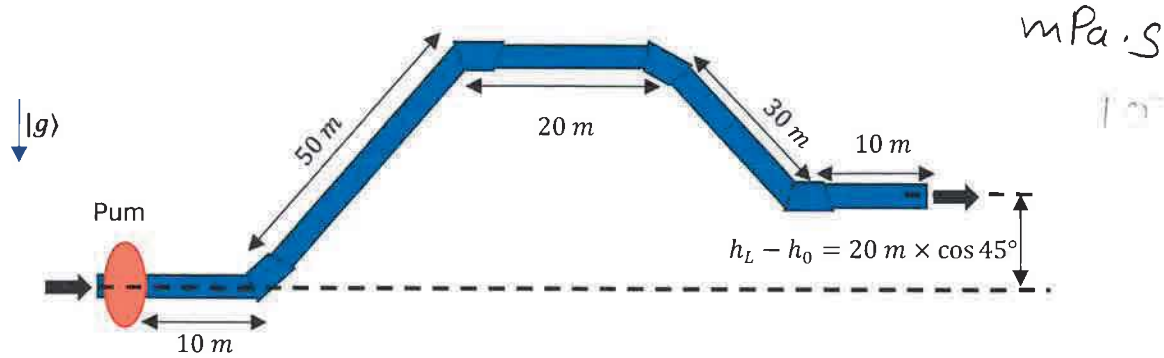
- 1.2. Using the equation of motion, derive an ordinary differential equation (ODE) for  $v_\theta$ . Integrate the ODE to get an equation for  $v_\theta$  as a function of  $r$ . Write down two boundary conditions for  $v_\theta$  and evaluate the integration constants.

15/30

10/20

Name:

2. Water is pumped between two reservoirs at the rate of  $10 \frac{\text{kg}}{\text{s}}$ . The pipeline consists of a pump, five smooth circular pipes of 20 cm inner radius (lengths of which are provided in the schematic below), and four 45° elbows. The resistance of each elbow is roughly equivalent to that offered by a pipe whose length is 15 diameters. The density of water is  $\rho = 1000 \frac{\text{kg}}{\text{m}^3}$  and its viscosity is  $\mu = 1.06 \text{ mPa} \cdot \text{s}$ .



- 2.1. Using the provided chart, determine the Fanning friction factor for the above system.
- 2.2. If both reservoirs are at atmospheric pressure ( $1.01 \times 10^5 \text{ Pa}$ ), what is the pressure at the pump outlet?

$$Re = \frac{\rho \langle v \rangle D}{\mu}$$

$$F_k \equiv (2\pi RL) \left( \frac{1}{2} \rho \langle v \rangle^2 \right) f = (\mathcal{P}_0 - \mathcal{P}_L) \pi R^2$$

Name:

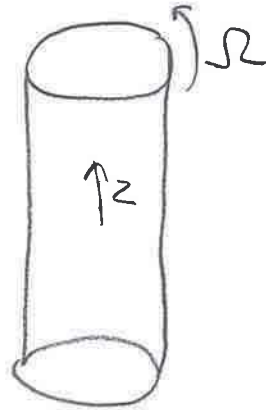
Parthiv Patel EOC

$$1.1 \quad \cancel{\frac{\partial p}{\partial t}} + \frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{1}{z} \frac{\partial}{\partial \theta} (r v_\theta) + \frac{\partial}{\partial z} (r v_z) = 0$$

given: steady state laminar flow  
 $\mu$  &  $\rho$  are constant

$$v_r = 0 \text{ and } v_\theta = 0$$

$$\frac{\partial p}{\partial \theta} = 0$$



If we simplify the above equation

$$\frac{1}{z} \frac{\partial}{\partial \theta} (r v_\theta) = 0$$

$$\text{Thus } \boxed{v_\theta = 0}$$

$\Rightarrow v_\theta$  does not change in  $\theta$  direction.

1.2

$$0 = \mu \left[ \frac{\partial}{\partial r} \left( \frac{1}{r} \frac{\partial}{\partial r} (r v_z) \right) \right] + \rho g_\theta$$

$$\frac{\partial}{\partial r} \left( \frac{\partial}{\partial r} (r v_z) \right) = \int - \frac{\rho g_\theta r}{\mu}$$

$$\frac{\partial}{\partial r} (r v_z) = \frac{-\rho g_\theta r^2}{2\mu} + C_1$$

$\Rightarrow$  we know that at  $r=0$   $v_z=0$  BC.

$$\text{Thus } \boxed{C_1 = 0}$$

Name:

$$\int \frac{d}{dr} (r V_0) = \int \frac{-\rho_0 r^2}{2\mu}$$

$$r V_0 = -\frac{\rho_0 r^3}{6\mu} + C_2$$

$$\Rightarrow \text{at } V_0 = 0 \quad r = R \quad \text{BC.}$$

$$V_0 = -\frac{\rho_0 R^3}{6\mu} + C_2$$

$$C_2 = +\frac{\rho_0 R^3}{6\mu}$$

$$V_0 = -\frac{\rho_0 r^3}{6\mu} + \frac{\rho_0 R^3}{6\mu}$$

$$V_0 = \frac{\rho_0}{6\mu} (R^3 - r^3)$$

$$r = R\Omega$$

$$V_0 = \frac{\rho_0}{6\mu} (R^3 - \Omega^3 R^3)$$

$$V_0 = \frac{\rho_0 R^3}{6\mu} (1 - \Omega^3)$$

$$1 \text{ kg} = 1 \text{ m}^3$$

Name:

$$\textcircled{2} \quad Q = \frac{10 \text{ m}^3}{\text{s}} \text{ in} = \frac{10 \text{ kg}}{\text{s}}$$

(1')

Four  $45^\circ$  elbows are equivalent to 1 SD of length = 6 m

$$L_{\text{net}} = 120 \text{ m} + 0.4 \text{ m} + 6 \text{ m} = 126.4 \text{ m}$$

$$D = 2 \times 20 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ cm}} = 0.4 \text{ m}$$

$$R = 0.2 \text{ m}$$

$$\langle v \rangle = \frac{Q}{A}$$

$$\langle v \rangle = \frac{10 \text{ m}^3}{\text{s}}$$

$$\frac{10}{2 \times 3.14 \times 0.2^2 \times 126.4 \text{ m}} = 0.06298879 \frac{\text{m}}{\text{s}}$$

$$Re = \frac{\rho \langle v \rangle D}{\mu} = 1100$$

$$= \frac{1000 \frac{\text{kg}}{\text{m}^3} \times 0.06298879 \frac{\text{m}}{\text{s}} \times 0.4 \text{ m}}{1.06 \times 10^{-3} \text{ Pa} \cdot \text{s}} = 23769.35$$

$Re > 23769.35$  So not laminar.

$$f = \frac{0.0791}{Re^{1/4}} = \boxed{6.37 \times 10^{-3}}$$

Name:

$$(ii) F_K = (P_0 - P_L) \pi R^2$$

$$F_K = (2\pi RL) \left( \frac{1}{2} \rho \langle v^2 \rangle \right) f$$

$$= (4 \times 3.14 \times 0.2_m \times 126.4_m) \times \left( \frac{1}{2} \times 1000 \frac{kg}{m^3} \times (0.063 \frac{m}{s})^2 \right) \times 6.37 \times 10^{-3}$$

$$F_K = 0.202935878$$

$$\frac{(0.202935878 \frac{kg}{s})}{3.14 \times (0.2_m)^2} \Rightarrow P_0 - P_L$$

$$P_0 - P_L = 1.6668 \text{ Pa}$$

$$P_{\text{rel}} = (P_0 - P_L) + P_{\text{atm}} + P_{\text{relative}}$$

$$1.6668 + 2(1.013 \times 10^5) + 9.81 - 70_m \times (0.925)$$
$$= 1.6668 + 2(1.013 \times 10^5) = 138,575.191$$

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